

OFRA and ETG Technical Documentation

Current five-dataset protocol and evidence | 1 September 2026

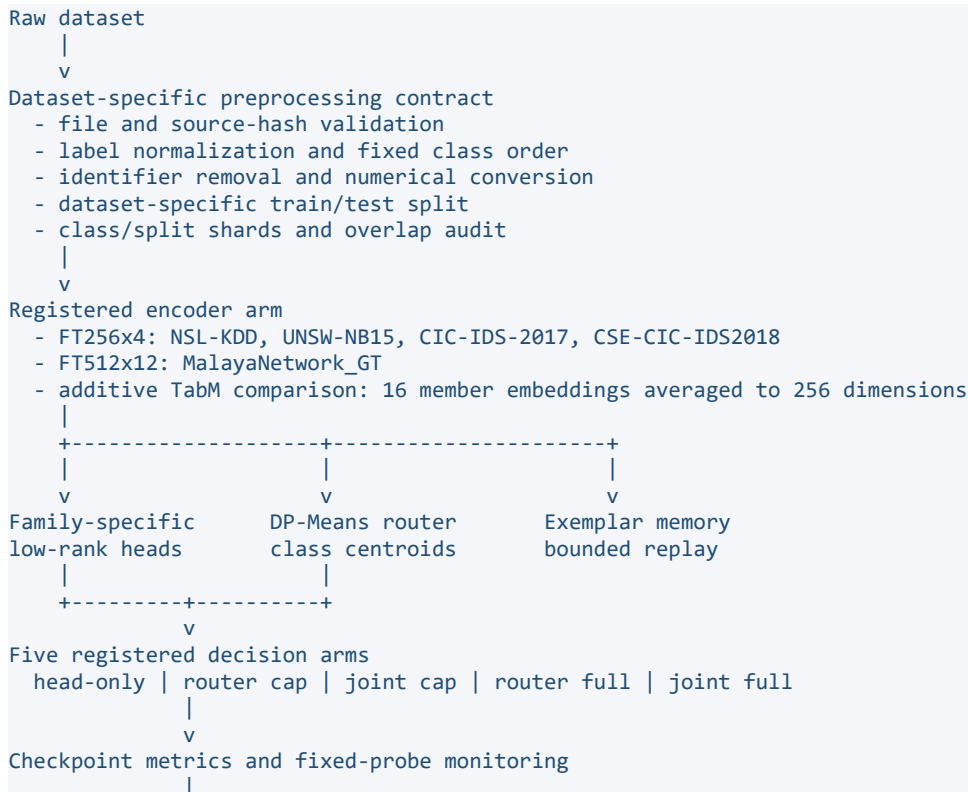
1. Research objective and evaluation boundary

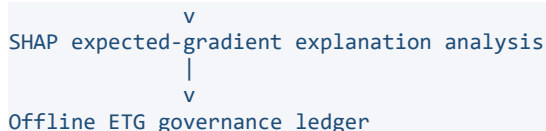
The project studies class-incremental learning for network-traffic classification. A class-incremental learner receives an ordered sequence of tasks. Each task introduces classes that were not available in earlier tasks, and evaluation covers all classes seen so far. The practical problem is to add new traffic classes without retraining from scratch while limiting catastrophic forgetting of earlier classes.

Four datasets are intrusion-detection benchmarks: NSL-KDD, UNSW-NB15, CIC-IDS-2017, and CSE-CIC-IDS2018. MalayaNetwork_GT is an external application-traffic dataset with application/service labels. It tests the same class-incremental mechanism under non-IID capture conditions but is not treated as an intrusion-detection benchmark.

Each dataset is processed, trained, and evaluated independently. Rows from different datasets are never pooled into a shared training stream.

2. End-to-end architecture





The classifier produces the prediction before SHAP and ETG are run. SHAP explains the stored decision evidence. ETG records whether the explanation and performance evidence satisfy a registered governance rule. ETG does not alter the class predicted by the model in the published experiment.

3. Dataset-specific preprocessing contracts

The current strict cache manifests and bound result artifacts are the source of truth for the published rows. The builders operate in chunks, validate the required files and columns, reject unexpected labels in strict mode, convert features to finite numerical arrays, and write class-by-split shards. Every current cache records input and output SHA-256 hashes, feature order, class order, task schedule, row accounting, split-overlap audit, and the Task-0-only preprocessing state. `fullcache/specs.py` remains as a traceable legacy builder for the earlier 122/194-feature contracts and must not be used to describe the current strict NSL-KDD or UNSW-NB15 results.

Dataset	Input and split	Removed fields	Output and task schedule
NSL-KDD	KDDTrain+.txt and KDDTest+.txt; official split retained; protocol, service, and flag vocabularies are fitted on Task-0 training rows and frozen; later unseen values use all-zero categorical blocks	label and difficulty	116 model features; classes Normal, DoS, Probe, R2L, U2R; tasks (Normal, DoS) -> Probe -> R2L -> U2R
UNSW-NB15	official 175,341-row training and 82,332-row testing CSVs retained; protocol, service, and state vocabularies are fitted on Task-0 training rows and frozen; later unseen values use all-zero categorical blocks	id, binary label, attack category	186 model features; ten classes; five two-class tasks beginning with Normal and Generic
CIC-IDS-2017	eight labelled flow CSVs; deterministic 80/20 grouping by cleaned feature bytes	label, flow ID, source/destination IP, source port, timestamp, external IP	78 features; eight grouped classes; four two-class tasks
CSE-CIC-IDS2018	all ten official labelled traffic CSVs; deterministic 80/20 grouping by cleaned feature bytes	label, timestamp, flow ID, source IP/port, destination IP; destination port is retained	78 features; seven grouped classes; four tasks
MalayaNet work_GT	31 derived-flow CSVs at frozen revision 384a59278f98490ee6e93aae017e748078d29b6a; one frozen capture per class is held out	source/destination IP, source/destination port, timestamp	77 numerical flow features; ten application classes; five two-class tasks

For the feature-hash split, identical cleaned feature rows share the same split assignment, preventing an exact cleaned duplicate from appearing in both train and test. Malaya uses a capture-level split instead because rows from the same capture are correlated; a row-level random split would leak capture-specific structure into evaluation.

3.1 No-look-ahead preprocessing audit

The numerical normalization path passes a strict temporal audit. The `FrozenTask0Stats` accumulator reads only Task-0 training shards, is frozen before Task-0 pretraining, and rejects later updates. It uses `float64` population variance and never reads official-test rows.

The current categorical path also passes the registered audit. NSL-KDD and UNSW-NB15 fit their one-hot vocabularies on Task-0 training rows only and freeze those vocabularies before the class-incremental stream proceeds. A categorical value that first appears in a later task is mapped to the pre-registered all-zero block for that field. The resulting strict widths are 116 for NSL-KDD and 186 for UNSW-NB15. The bound current caches record 133 NSL-KDD rows and 953 UNSW-NB15 rows that use this unknown-category handling.

The earlier audit, retained only as historical motivation, found that the legacy complete-training-partition vocabulary would have introduced:

- NSL-KDD contains six future-only columns (service: aol, harvest, http_2784, http_8001, pm_dump; flag: RSTOS0). They affect 115 of 12,703 later-task rows (0.905%).
- UNSW-NB15 contains eight future-only columns (proto: argus, bbn-rcc, crtp, egg, hmp, netblt, rdp; service: irc). They affect 712 of 79,341 later-task rows (0.897%).

That legacy behaviour was bounded transductive schema information rather than official-test leakage or future-label training. It is not the preprocessing contract used by the current formal prediction results. The strict Task-0-only caches were rebuilt and the affected five-seed experiments were rerun. CIC-IDS-2017, CSE-CIC-IDS2018, and MalayaNetwork_GT use numerical model inputs and have no data-derived categorical vocabulary.

4. Model and continual-learning components

4.1 FT-Transformer encoder

The Malaya formal configuration uses a 512-dimensional FT-Transformer with 12 layers, 16 attention heads, head dimension 32, and 0.1 attention/feed-forward dropout. Its initial task is trained for eight epochs and each later task for ten epochs. The current NSL-KDD, UNSW-NB15, CIC-IDS-2017, and CSE-CIC-IDS2018 formal rows use the smaller FT256x4 protocol recorded in their bound artifacts; the strict CSE-CIC-IDS2018 result also uses the 8+10-epoch schedule. Model capacity is therefore disclosed per dataset and is not silently pooled. In every case the encoder maps a numerical traffic row x to representation $h(x)$ and is shared by all classes within that dataset run.

An epoch is one pass over the training data selected for that stage. More epochs provide more optimization steps but can also increase overfitting or forgetting; all epoch counts are protocol settings, not universal optima.

4.2 Additive TabM encoder comparison

TabM is evaluated as an additional encoder arm rather than a replacement for the registered OFRA method. The adapter produces 16 member embeddings with shape (batch, 16, 256) and averages

them across the member dimension to obtain the single (batch, 256) representation required by the existing OFRA interface. The family heads, exemplar budget, DP-Means routers, joint-score weight, task schedule, and evaluation definitions remain unchanged.

This is a deliberately narrow integration. It tests whether a different tabular representation improves the existing prediction pipeline without introducing a new router or governance rule. It must be described as a mean-embedding TabM adapter, not as a full member-wise redesign of OFRA.

4.3 Family-specific low-rank heads

Each seen class has a binary positive-versus-negative decision head. The head uses a rank-8 low-rank adaptation with scaling parameter 16. For class c , the head outputs probability $p(c, x)$. Family-specific heads allow a new class to receive a new decision component without rebuilding a single fixed multiclass output layer.

4.4 Bounded exemplar memory

The learner retains at most 50 exemplars per class from a candidate pool of at most 5,000. Replay mixes selected older examples with the current task to reduce catastrophic forgetting. This is a fixed study budget and does not imply that 50 is generally optimal.

4.5 DP-Means router

The router maintains one or more centroids for each seen class in encoder space. DP-Means can create an additional centroid when an embedding is sufficiently far from the existing centroids, subject to a maximum of 32 centroids. The creation threshold is derived from the registered 0.3 distance quantile.

For class c , the raw routing score is the negative distance from $h(x)$ to the nearest centroid. Scores are standardized across seen classes to obtain $z(c, x)$. A higher value means the sample is more compatible with that class's stored embedding geometry.

The capped router fits centroids from at most 3,000 selected samples per class; the uncapped router uses all eligible samples. The cap controls router fitting cost and prevents large classes from dominating centroid estimation. It does not limit the number of test rows.

4.6 Registered decision arms

The experiment evaluates matched views of the same trained model:

- head-only: $s(c, x) = p(c, x)$;
- router cap: $s(c, x) = z_cap(c, x)$;
- joint cap: $s(c, x) = p(c, x) + 0.5 \cdot z_cap(c, x)$;
- router full: $s(c, x) = z_full(c, x)$;

- joint full: $s(c, x) = p(c, x) + 0.5 \cdot z_{\text{full}}(c, x)$.

The predicted class is $\text{argmax}_c s(c, x)$, meaning the class with the largest score among all classes seen at that checkpoint. The joint weight 0.5, cap 3,000, quantile 0.3, and centroid limit 32 are registered project settings.

4.7 CatBoost cumulative diagnostic

CatBoost is used only as a cumulative multiclass diagnostic. At each checkpoint it trains a fresh classifier on all training rows belonging to the classes seen so far. It therefore estimates classification headroom in the fixed Malaya feature representation, but it does not use OFRA family heads, bounded exemplar replay, DP-Means routing, or the cap-3,000 budget. Its native feature attribution is not the registered routed-margin SHAP target and is not entered into the ETG ledger. A CatBoost-OFRA hybrid would constitute a new method and is outside the frozen protocol.

5. Training and evaluation protocol

The formal configuration uses focal loss with gamma 2.0 and alpha 0.75, learning rate 0.001, batch size 384, evaluation batch size 512, and a negative-to-positive sampling ratio of 4. Deterministic execution and shard hash verification are enabled. Recovery checkpoints are validated by hash before a run resumes.

At checkpoint t , the model is evaluated on every class seen through task t . Reported metrics include:

- overall accuracy: fraction of correct predictions;
- Macro-F1: arithmetic mean of per-class F1, giving every class equal weight;
- balanced accuracy: arithmetic mean of per-class recall;
- average forgetting: mean decline from each earlier task's best historical

accuracy to its current accuracy, reported in percentage points;

- benign false-positive rate and attack-detection recall for datasets with a

valid benign/attack interpretation.

MalayaNetwork_GT does not report benign/attack metrics because its labels are applications, not attack categories.

6. SHAP explanation analysis

SHAP is a family of additive feature-attribution methods. The current analyzer uses `shap.GradientExplainer`, an expected-gradients approximation suitable for the differentiable FT-Transformer. It does not use KernelSHAP.

For a fixed probe and checkpoint, the analyzer reconstructs the registered `joint_cap3000` class margin and attributes that margin to input features. A positive attribution pushes the explained margin upward; a negative value pushes it downward. The largest absolute values identify the features with the strongest contribution for that particular probe and decision. The commonly shown top-15 list is therefore ranked by absolute SHAP magnitude, not selected manually.

Explanation drift is measured across eligible class-by-adjacent-checkpoint transitions. Feature-ranking overlap uses Jaccard similarity:

$$J(A,B) = |A \text{ intersection } B| / |A \text{ union } B|.$$

Here A and B are the selected top-feature sets at two adjacent checkpoints. A silent explanation-drift event is a registered condition in which predictive performance remains within its stability rule while the explanation changes beyond the explanation threshold. The thresholds, including a 0.7 rule used in the registered analysis, are study-defined governance settings and are not claimed as universal standards.

6.1 Attribution-method robustness

The registered explanation is Expected Gradients. A source-bound Malaya seed-1 pilot also evaluates two alternative rankings against the same frozen `joint_cap3000` class margin, probes, checkpoints, top-15 rule, random-control admission threshold, and ETG state machine:

- single-feature ablation replaces one input at a time with the frozen checkpoint mean and measures the resulting margin decrease;
- Gradient x Input multiplies each raw input by the local margin gradient.

Across the 30 common checkpoint-class rows, all three primary methods agree on the admission decision for 40.0% and the ETG state for 43.3%. Across the 20 common adjacent class transitions, all three agree on the silent-drift conclusion for 20.0%. Pairwise mean top-15 Jaccard is 0.567 for Expected Gradients versus feature ablation, 0.380 for Expected Gradients versus Gradient x Input, and 0.405 for feature ablation versus Gradient x Input. The associated silent-drift counts are 12/17, 0/17, and 7/17, respectively. ETG outcomes are therefore method-conditioned.

Integrated Gradients were also attempted with the mean Task-0 background and 16-point Gauss-Legendre quadrature. Its completeness check failed on this piecewise routed score (mean row-level error 11.58; maximum 373.72), so it is retained as a numerical diagnostic and excluded from primary agreement claims. This failure is evidence that an attribution method suitable for a smooth encoder is not automatically valid for the complete routed scorer.

7. ETG: Explanation Trust Governance

ETG is the project's offline explanation-governance layer. Its evidence unit is a class-by-adjacent-checkpoint transition, not a packet, flow, sample, user, or real-world incident. Each ledger row links the earlier checkpoint, later checkpoint, performance evidence, explanation evidence, registered thresholds, and resulting governance state.

The published strict ETG analysis can:

- certify an admission when the registered evidence is acceptable;
- refuse admission when evidence is insufficient or violates a rule;
- escalate a transition for further review;
- require strict recertification after drift;
- record whether that recertification passed or failed.

These actions are simulated governance outcomes. They are not evidence that a human review, deployment block, or production remediation actually occurred. The completed ledger consumes stored OFRA checkpoint evidence; it does not yet demonstrate a closed-loop causal system in which ETG changes future OFRA routing or training.

8. Current completed evidence

8.1 Current five-seed prediction evidence

The current strict table uses seeds 1, 2, 3, 4, 42. Each dataset was trained and evaluated independently. NSL-KDD, UNSW-NB15, CIC-IDS-2017, and CSE-CIC-IDS2018 use the FT-Transformer 256x4 protocol; MalayaNetwork_GT uses FT-Transformer 512x12. CSE-CIC-IDS2018 uses the registered 8+10-epoch schedule but remains a separately interpreted failure-mode result because its scale and class imbalance differ sharply from the other datasets. The table is therefore a dataset-by-dataset audit, not a capacity-matched pooled superiority test.

Dataset	Model	Arm	Accuracy	Macro-F1	Balanced accuracy	Forgetting
NSL-KDD	FT256x4	Head only	56.29%	31.08%	36.48%	8.14 pp
NSL-KDD	FT256x4	Joint cap 3,000	71.56%	41.46%	42.25%	1.60 pp
UNSW-NB15	FT256x4	Head only	68.78%	21.16%	23.52%	2.68 pp
UNSW-NB15	FT256x4	Joint cap 3,000	61.95%	23.50%	28.45%	8.97 pp
CIC-IDS-2017	FT256x4	Head only	64.59%	21.25%	25.91%	9.04 pp
CIC-IDS-2017	FT256x4	Joint cap 3,000	72.57%	36.95%	63.13%	9.63 pp
CSE-CIC-IDS2018	FT256x4	Head only	79.21%	19.06%	21.15%	2.00 pp
CSE-CIC-IDS2018	FT256x4	Joint cap 3,000	50.16%	33.55%	52.15%	19.07 pp

Dataset	Model	Arm	Accuracy	Macro-F1	Balanced accuracy	Forgetting
MalayaNetwork_GT	FT512x12	Head only	56.03%	11.77%	15.05%	3.16 pp
MalayaNetwork_GT	FT512x12	Joint cap 3,000	54.68%	21.15%	23.03%	3.55 pp

The results are heterogeneous. Joint cap 3,000 improves all four displayed metrics on NSL-KDD. On UNSW-NB15 it improves class-balanced metrics but lowers overall accuracy and increases forgetting. On CIC-IDS-2017 it improves accuracy and class-balanced metrics but does not reduce forgetting. On CSE-CIC-IDS2018 it substantially improves class-balanced detection and attack recall but imposes severe accuracy, false-positive, and forgetting costs. On Malaya it improves minority-class coverage while its accuracy and forgetting effects remain uncertain. No universal-superiority claim is supported by this table.

For Malaya FT512x12, the mean +/- sample standard deviation is:

Arm	Accuracy	Macro-F1	Balanced accuracy	Forgetting
Head only	56.03% +/- 0.45	11.77% +/- 0.81	15.05% +/- 0.55	3.16 +/- 4.39 pp
Joint cap 3,000	54.68% +/- 2.71	21.15% +/- 3.37	23.03% +/- 3.42	3.55 +/- 0.77 pp

The paired joint-minus-head mean differences are -1.34 percentage points for accuracy (95% CI -4.71 to +2.02), +9.37 points for Macro-F1 (95% CI +4.70 to +14.04), +7.97 points for balanced accuracy (95% CI +3.72 to +12.23), and +0.39 points for forgetting (95% CI -5.75 to +6.54; negative would favour the joint arm). The result supports a class-balance improvement, not a stable forgetting reduction. Pooled per-class results further show that joint cap 3,000 produces non-zero recall for every Malaya class, while head-only has zero recall for six of ten classes; several minority recalls nevertheless remain low.

8.2 Explanation drift and offline ETG evidence

The completed source-bound explanation analysis currently covers Malaya seed 1. The registered primary rule uses Expected Gradients, the actual joint_cap3000 class margin, top-15 features, a Jaccard drift threshold of 0.7, and an allowed class-recall drop of 5 percentage points. It reports 12 silent explanation-drift events among 17 eligible class-by-adjacent-checkpoint transitions (70.59%). This is not a packet, flow, sample, or production-incident rate.

ETG records six certified admissions, four refused admissions, four escalations, one strict recertification, and two strict-recertification failures. These are simulated offline governance outcomes over stored OFRA evidence. They do not mean that a human review occurred or that ETG changed future routing, replay, training, or predictions.

The sensitivity grid varies top-k over {10, 15, 20}, the Jaccard threshold over {0.6, 0.7, 0.8}, and the allowed class-recall drop over {2, 5, 10} percentage points. It confirms that the reported rate is threshold-sensitive. The third dimension must not be called an overall-accuracy tolerance. Because the

grid is single-seed, it bounds the registered decision rule but does not identify a universal operating threshold.

The attribution-method robustness pilot is also bounded to Malaya seed 1. Expected Gradients, single-feature ablation, and Gradient x Input agree on all admission decisions for 40.0% of the evaluated cases, on ETG state for 43.3%, and on the silent-drift conclusion for 20.0%. Integrated Gradients failed the recorded completeness diagnostic and is excluded from the primary comparison. The governance conclusion is therefore method-sensitive and must be reported with this limitation.

8.3 Additive classifier comparisons on MalayaNetwork_GT

The TabM mean-embedding adapter has completed the full OFRA prediction pipeline for seeds 1, 2, 3, 4, 42. The table reports the official test view; all values are mean +/- sample standard deviation.

TabM-OFRA arm	Accuracy	Macro-F1	Balanced accuracy	Forgetting
Head only	56.17% +/- 1.79	11.11% +/- 1.99	14.43% +/- 2.35	0.61 +/- 0.46 pp
Router only, cap 3,000	40.15% +/- 3.49	17.44% +/- 1.44	19.29% +/- 1.11	12.75 +/- 1.89 pp
Joint, cap 3,000	57.99% +/- 3.93	20.46% +/- 1.19	21.60% +/- 1.70	4.39 +/- 2.56 pp
Joint, uncapped	57.93% +/- 3.84	20.34% +/- 1.31	21.48% +/- 1.87	4.33 +/- 2.39 pp

The earlier four-common-seed FT-versus-TabM diagnostic remains an additive classifier comparison rather than a primary OFRA claim. It suggested that the TabM gain was mainly aggregate accuracy rather than broad class-balanced improvement. The now-complete five-seed FT512x12 result is the authoritative FT prediction record; the historical four-seed comparison remains useful only for tracing how the capacity experiment was developed.

Matched cumulative multiclass diagnostics over seeds 1, 2, 3, 4, 42 give CatBoost 66.50% +/- 0.44 accuracy, 34.76% +/- 0.69 Macro-F1, 39.10% +/- 0.73 balanced accuracy, and 15.34 +/- 0.73 points of forgetting. The corresponding cumulative TabM values are 65.01% +/- 0.96, 32.66% +/- 1.31, 34.41% +/- 1.51, and 9.43 +/- 1.80 points. These cumulative results are classifier-capacity diagnostics and are not matched OFRA comparisons.

8.4 CSE-CIC-IDS2018 strict failure-mode result

The strict FT256x4 campaign completed seeds 1, 2, 3, 4, and 42 with eight Task-0 epochs and ten epochs per later task as DICC Job 402073. The protected output manifest and all five seed files passed SHA-256 verification.

Arm	Accuracy	Macro-F1	Balanced accuracy	Forgetting	Attack recall	Benign FPR
Head only	79.21%	19.06%	21.15%	2.00 pp	31.70%	6.57%
Joint, cap 3,000	50.16%	33.55%	52.15%	19.07 pp	91.82%	51.85%

Arm	Accuracy	Macro-F1	Balanced accuracy	Forgetting	Attack recall	Benign FPR
Joint, uncapped	54.39%	33.72%	51.35%	15.39 pp	84.38%	46.48%

Relative to head-only inference, joint cap-3,000 increases Macro-F1 by 14.48 points and attack recall by 60.12 points, but reduces overall accuracy by 29.05 points, increases forgetting by 17.07 points, and increases benign false-positive rate by 45.29 points. The uncapped joint arm reduces but does not remove these operating costs. This is evidence of a severe detection coverage trade-off, not a deployable improvement. The older FT256x4 1+1-epoch closure remains historical evidence and is not mixed with this current result.

8.5 Open-set and labelled-head diagnostic

The evaluator used the checkpoint immediately before FTP-Patator was introduced as a labelled task. Its thresholds came from 11,647 training-only known-class calibration rows, and the held-out test class was not used to tune them. All four registered gates produced 0% unknown recall. The best continuous AUROC was 0.5303, the OSCR-style AUC was 0.4699, and the primary candidate buffer contained 509 known rows and no FTP-Patator rows. The gate therefore failed the discovery objective.

Once FTP-Patator labels were supplied, OFRA used its ordinary supervised class-incremental mechanism to create the corresponding head. New-class recall was 82.30%; old-class accuracy moved from 87.28% to 87.01%, a 0.27-percentage-point decrease. The evaluator did not create a new head automatically. This is a single-seed expected-contract diagnostic, so the next step is to redesign the gate and rotate held-out classes before a multi-seed claim.

Quantity	Verified seed-42 result
Known / unknown test rows	226,135 / 1,588
Unknown recall (all gates)	0.00%
Primary known false-unknown rate	0.225%
Best AUROC / OSCR-style AUC	0.5303 / 0.4699
Post-label new-class recall	82.30%
Old-class accuracy change	-0.27 percentage points

8.6 Paired checkpoint-selection experiment

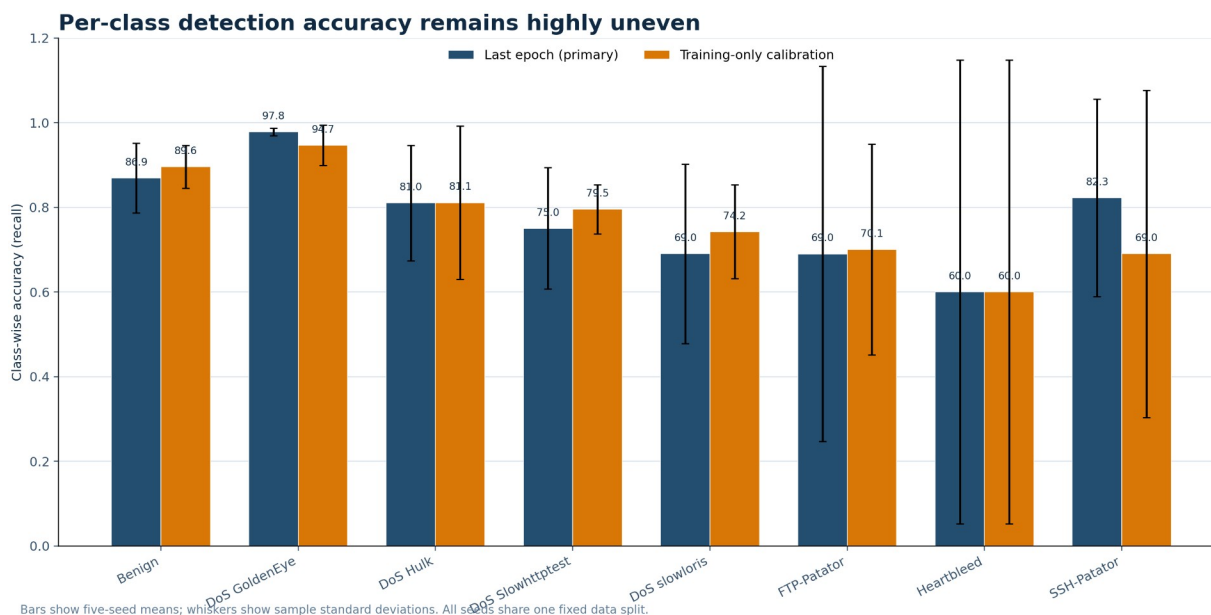
DICC Job 425539 completed ten matched runs: last-epoch and training-only calibration for seeds 1, 2, 3, 4, and 42. All ten result/protocol/summary records, the W&B registry, the aggregate report, and the protected checksum manifest passed SHA-256 verification. The two arms share one fixed data split, model, replay budget, task sequence, optimizer, and joint-cap-3,000 evaluation. Only the per-class checkpoint-selection rule changes.

Aggregate five-seed result. A positive paired delta means calibration minus last epoch; for forgetting and benign FPR, a negative delta is desirable.

Metric	Last epoch	Training-only calibration	Paired delta (95% CI)
Final accuracy	85.51% +/- 6.18%	87.52% +/- 4.78%	+2.02% (-1.16%, +5.19%)
Macro-F1	54.40% +/- 5.07%	56.34% +/- 4.06%	+1.95% (-3.50%, +7.39%)
Average task accuracy	79.78% +/- 6.86%	77.88% +/- 9.57%	-1.91% (-7.48%, +3.66%)
Forgetting	3.52% +/- 2.60%	5.18% +/- 5.43%	+1.66% (-3.53%, +6.84%)
Attack recall	85.73% +/- 14.98%	83.76% +/- 16.72%	-1.97% (-4.41%, +0.48%)
Benign FPR	13.10% +/- 8.25%	10.44% +/- 5.05%	-2.66% (-6.83%, +1.51%)

The candidate improves final accuracy by 2.02 percentage points, Macro-F1 by 1.95 points, and benign FPR by 2.66 points in the favourable direction. It simultaneously reduces average task accuracy by 1.91 points, increases forgetting by 1.66 points, and reduces attack recall by 1.97 points. Every paired 95% confidence interval crosses zero; the two primary outcomes have Holm-adjusted $p=0.755$. This is a mixed, statistically inconclusive result.

Figure 4. Per-class official joint-cap-3,000 recall. Whiskers are sample standard deviations across five seeds on one fixed split.



Per-class result. Class-wise accuracy is recall; support is the fixed official-test count.

Class	Support	Last recall	Cal. recall	Last F1	Cal. F1
Benign	174,421	86.9% +/- 8.3%	89.6% +/- 5.0%	90.7% +/- 4.4%	92.1% +/- 2.8%
DoS GoldenEye	2,059	97.8% +/- 0.9%	94.7% +/- 4.8%	90.2% +/- 7.3%	92.5% +/- 2.0%
DoS Hulk	46,215	81.0% +/- 13.7%	81.1% +/- 18.1%	80.6% +/- 8.2%	82.7% +/- 7.5%
DoS Slowhttptest	1,100	75.0% +/- 14.3%	79.5% +/- 5.8%	54.6% +/- 7.7%	53.8% +/- 8.0%
DoS slowloris	1,159	69.0% +/- 21.2%	74.2% +/- 11.0%	48.3% +/- 16.5%	49.6% +/- 10.5%
FTP-Patator	1,588	69.0% +/- 44.3%	70.1% +/- 24.9%	30.9% +/- 28.5%	51.5% +/- 19.9%
Heartbleed	2	60.0% +/- 54.8%	60.0% +/- 54.8%	9.4% +/- 13.3%	7.6% +/- 10.4%
SSH-Patator	1,179	82.3% +/- 23.3%	69.0% +/- 38.7%	30.4% +/- 14.7%	21.0% +/- 16.6%

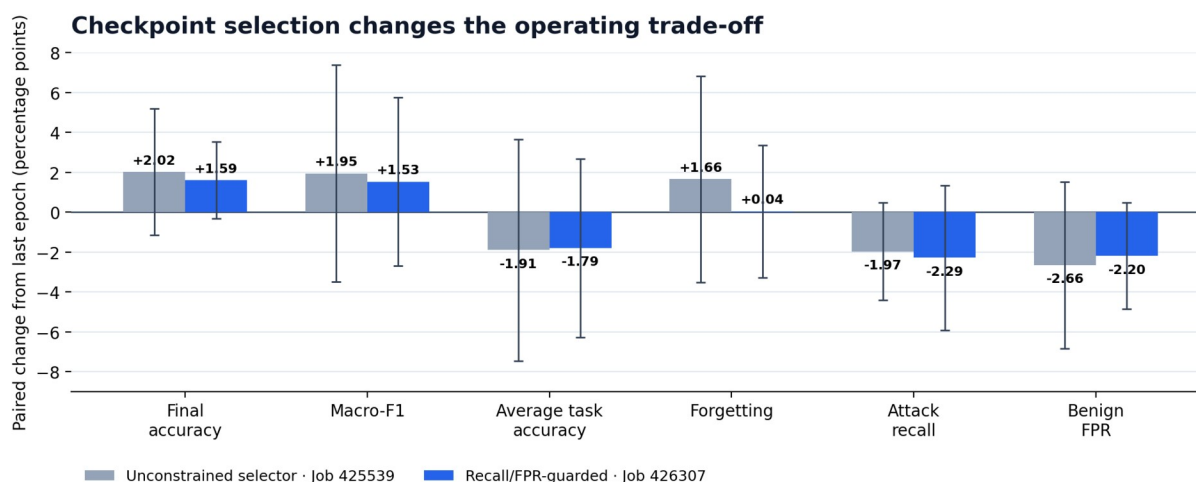
Interpretation: the calibration rule does not uniformly help the same classes. FTP-Patator F1 improves, but SSH-Patator recall and F1 decline. Heartbleed has only two test rows and only one training-calibration row, so it uses the registered fallback and cannot support a stable class-level claim. Last-epoch selection therefore remains the formal primary setting.

8.7 Recall/FPR-guarded checkpoint selection

The first training-only selector (Job 425539) chose the earliest maximum calibration Macro-F1 and improved final accuracy, Macro-F1, and benign FPR on average, but increased forgetting by 1.66 percentage points. Job 426307 tested a stricter project-defined gate: a non-final checkpoint is accepted only if calibration Macro-F1 improves by at least 0.01, attack recall falls by no more than 0.01, and benign FPR rises by no more than 0.01. These thresholds express this study's safety preference; they are not cited industry standards.

Across seeds 1, 2, 3, 4, and 42, guarded selection changed final accuracy by +1.59 points, Macro-F1 by +1.53, forgetting by +0.04, attack recall by -2.29, benign FPR by -2.20, and average task accuracy by -1.79 relative to matched last epoch. All paired 95% confidence intervals crossed zero. Macro-F1 had raw paired t $p=0.372$, Holm-adjusted $p=0.744$, and Wilcoxon $p=0.3125$; forgetting had $p=0.977$ and Wilcoxon $p=0.8125$. The guard recovers the unconstrained selector's mean forgetting penalty, but not a statistically supported overall win.

Figure 5. Unconstrained and guarded checkpoint selectors relative to the same last-epoch baseline.



Operational reading: the guard trades fewer benign false alarms for lower attack recall and does not improve every seed. Only seed 2 dominates its matched baseline across the five jointly inspected endpoint metrics. Per-class movements show error redistribution: for example, benign-to-FTP errors fall while benign-to-SSH errors rise. Heartbleed has two test rows and cannot support a population-level claim. The formal primary checkpoint remains epoch 10 / last epoch.

9. What is complete and what remains open

Completed in this release:

- MalayaNetwork_GT FT512x12 seeds 1, 2, 3, 4, 42;
- NSL-KDD, UNSW-NB15, and CIC-IDS-2017 FT256x4 five-seed formal results;
- CSE-CIC-IDS2018 FT256x4 strict 8+10-epoch five-seed result from DICC Job

402073, including verified seed files and protected result manifest;

- TabM mean-embedding OFRA prediction results on MalayaNetwork_GT for seeds 1, 2, 3, 4, 42;
- matched cumulative CatBoost and TabM diagnostics on MalayaNetwork_GT for the same five seeds;

- Malaya seed-1 SHAP and ETG analysis from completed DICC Job 389896;
- a source-bound three-method attribution robustness pilot, with the failed

Integrated-Gradients completeness diagnostic retained separately;

- a preprocessing no-look-ahead code/data audit;
- strict Task-0-only categorical-vocabulary caches and affected five-seed

reruns for NSL-KDD and UNSW-NB15;

- the historical protocol-separated CSE-CIC-IDS2018 FT256x4 1+1-epoch closure;
- CSE-CIC-IDS2018 A100 capacity evidence;
- executable preprocessing contracts for the five-dataset suite;
- deterministic result, source-binding, and publication hashes.
- seed-42 FTP-Patator open-set evaluation from DICC Job 414686, with 0% pre-label unknown recall and 82.30% post-label new-class recall;
- DICC Jobs 425539 and 426307: paired five-seed unconstrained and recall/FPR-guarded checkpoint-selection results, per-class evidence, W&B registrations, and protected SHA-256 manifests;

Not yet complete:

- SHAP and ETG analysis for the TabM arm; the current TabM result covers the

prediction pipeline only;

- multi-seed SHAP and ETG estimates;
- multi-seed attribution-method robustness estimates;
- protocol-matched external continual-learning baselines;
- inferential tests supporting superiority claims;
- a deployed feedback loop in which ETG decisions alter future OFRA routing or

training.

The current prediction table is a completed descriptive five-seed record for all five independently processed datasets. Multi-seed explanation/ETG, protocol-matched external baselines, and a deployed feedback loop remain open. The seed-42 FTP-Patator pilot is complete but negative for autonomous unknown discovery; only its post-label supervised adaptation succeeded. The ReplayIDS-aligned paired five-seed checkpoint-selection campaign is also complete and negative for a superiority claim; its per-class evidence is now included above. The guarded selector is complete as a five-seed mechanism ablation but is not the new primary setting because its confidence intervals include zero and attack recall decreases on average.

10. Reproducibility and file map

- `fullcache/`: preprocessing and cache verification;
- `streaming_full/`: current training, routing, evaluation, monitoring, and recovery;
- `ofra_encoders/`: FT-Transformer integration;
- `formal_v2_explanation_etg/`: SHAP and ETG analyzer;
- `results/`: per-seed JSON, current five-seed aggregate, paired and per-class tables, ETG records, sensitivity analysis, and capacity profile;
- `results/classifier_comparisons/`: bounded classifier-comparison summaries and their interpretation limits;
- `EXPERIMENT_SCOPE_FREEZE_20260813.md`: the frozen distinction between the primary method, additive comparisons, and changes requiring a new protocol;

- `reproducibility/`: exact runtime and analysis bindings;
- `SHA256SUMS.txt`: SHA-256 for every published file other than the manifest

itself.

The authoritative quantitative records are the JSON and CSV artifacts under `results/`. W&B is used to observe runs and compare logged metrics, but the repository artifacts and their hashes remain the reproducibility source of record.